

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
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Experiment 2: Binary Classification using Naïve Bayes and K-Nearest Neighbors

Aim

To build and evaluate Naïve Bayes and K-Nearest Neighbors (KNN) classifiers for a binary problem, analyze their performance using multiple metrics

Dataset Description

The Spambase dataset serves as a reference dataset for evaluating spam detection models. It is composed of numerical features extracted from email messages and a binary output label indicating spam or non-spam categories.

Dataset Reference:

- Kaggle – Spambase Dataset

Implementation

- Developed Naïve Bayes models using Gaussian, Multinomial, and Bernoulli variants
- Built an initial K-Nearest Neighbors (KNN) classifier as a baseline model
- Optimized KNN hyperparameters through GridSearchCV and RandomizedSearchCV techniques
- Evaluated the performance of KDTree and BallTree approaches for nearest-neighbor searches

Visualizations

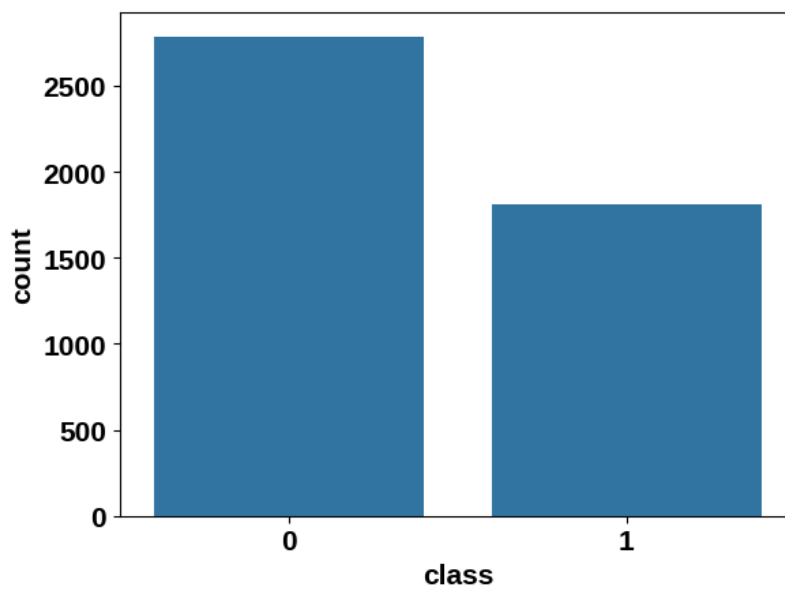


Figure 1: Class Distribution of Spam and Non-Spam Emails

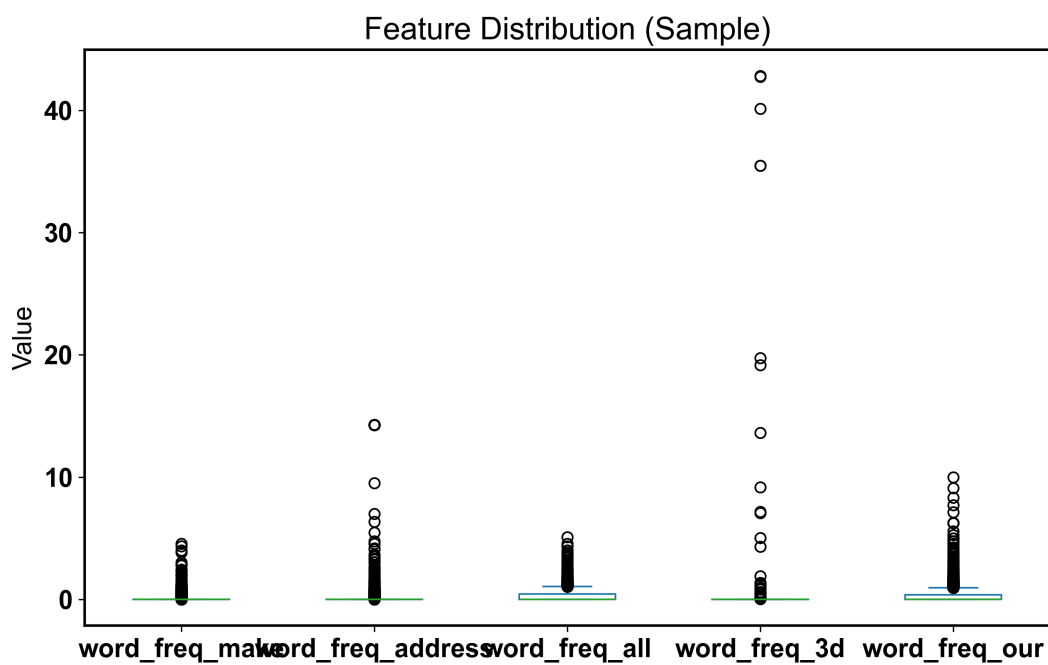


Figure 2: Feature Distribution Plot (Box plot)

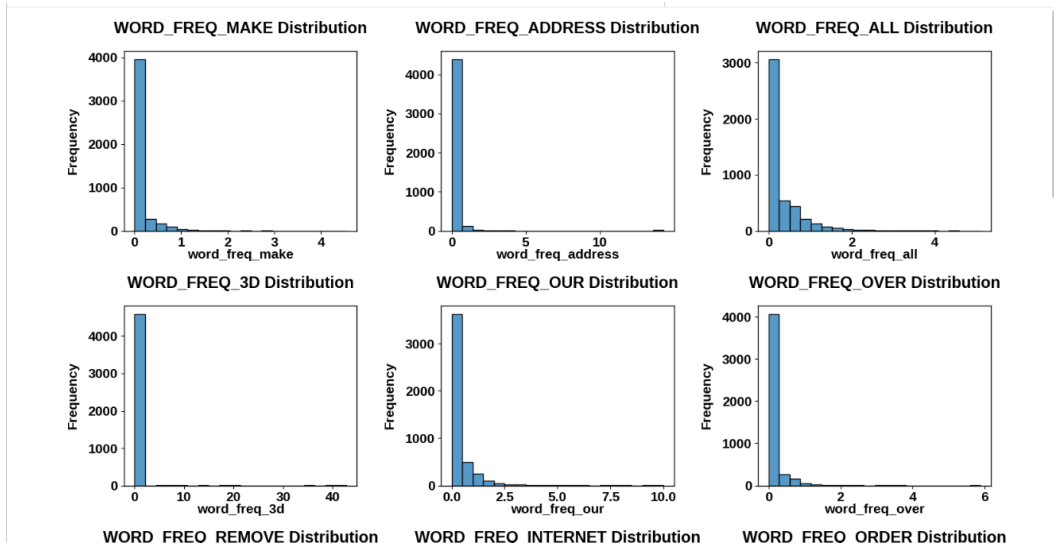


Figure 3: Feature Distribution Plot (Hist plot)

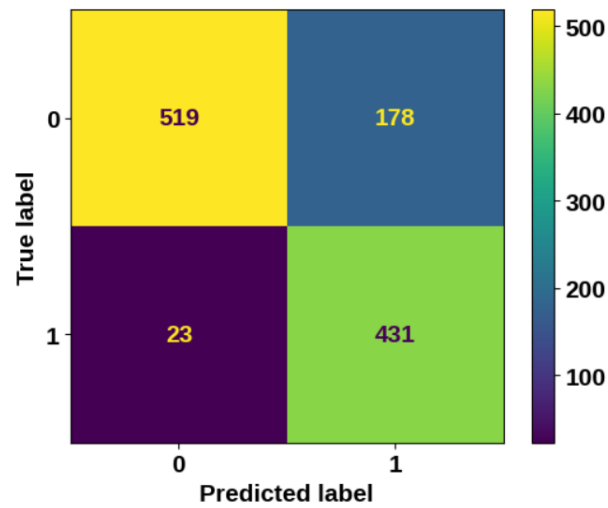


Figure 4: Confusion Matrix for Gaussian Naïve Bayes

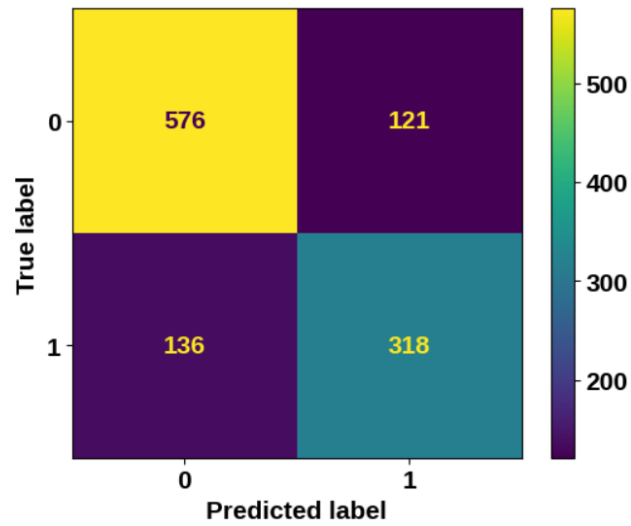


Figure 5: Confusion Matrix for Multinomial Naïve Bayes

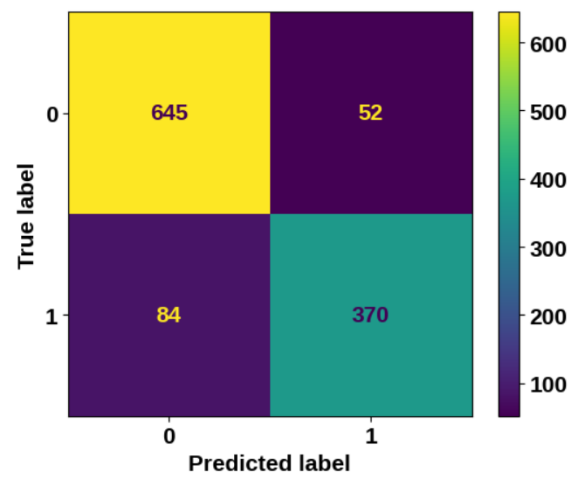


Figure 6: Confusion Matrix for Bernoulli Naïve Bayes

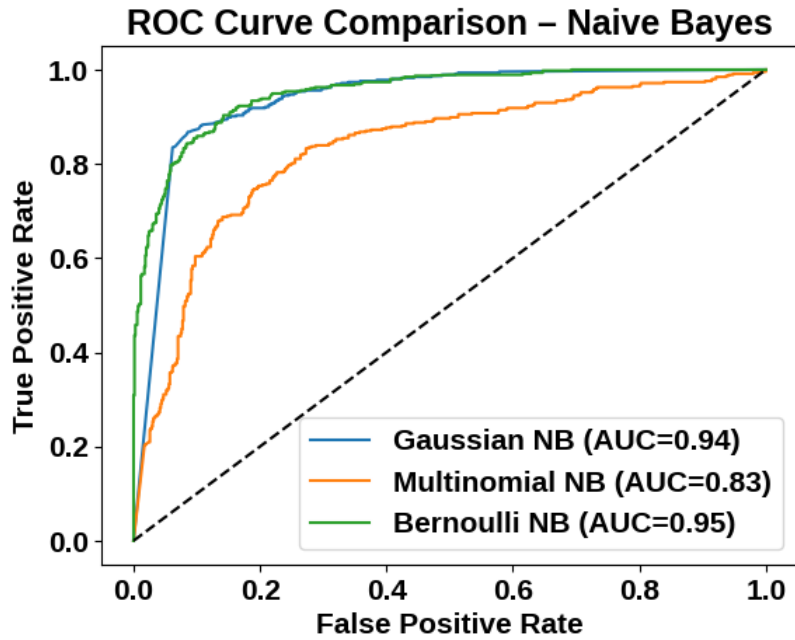


Figure 7: ROC Curve for Naïve Bayes

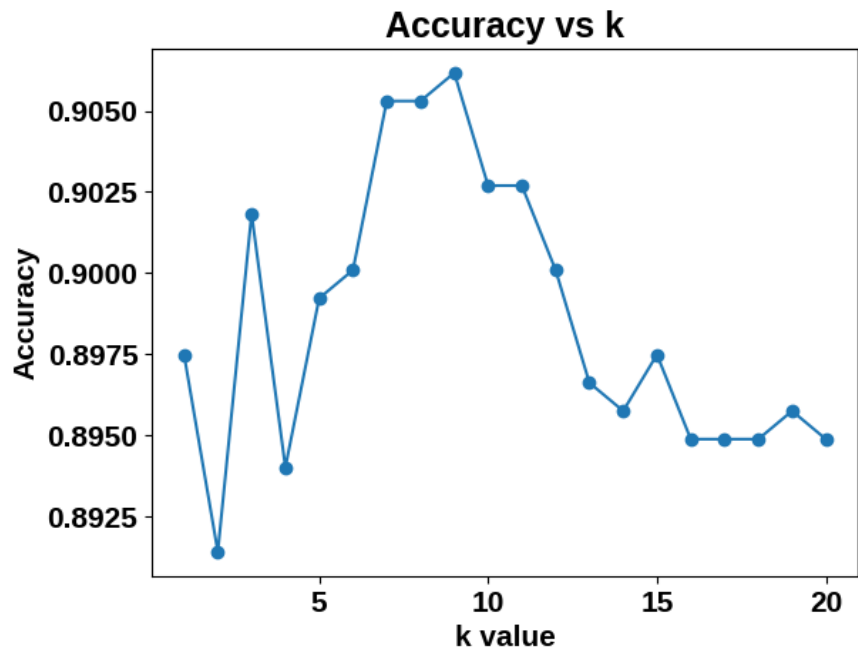


Figure 8: Accuracy vs. k Plot for KNN

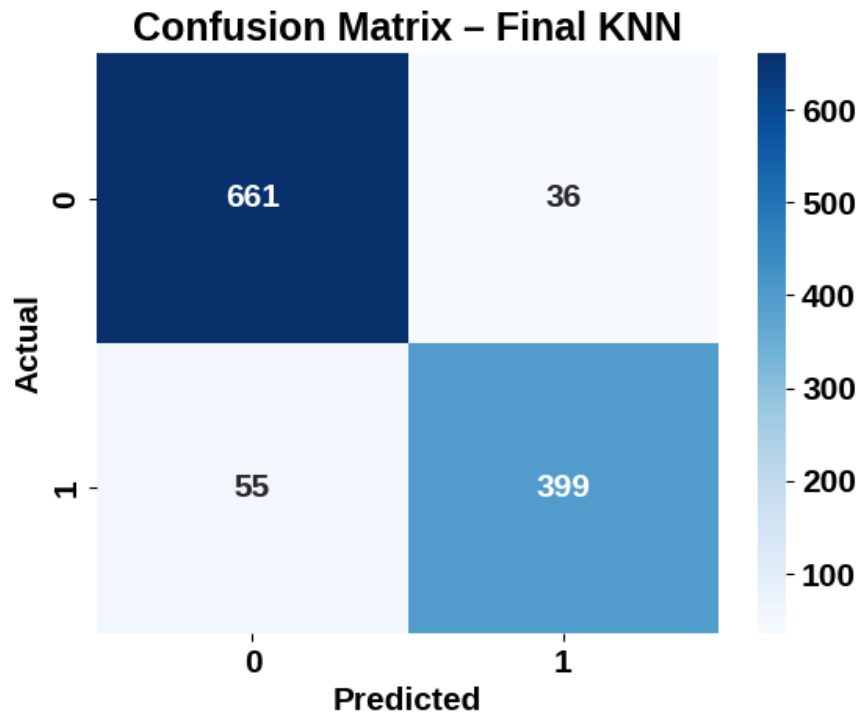


Figure 9: Confusion Matrix for Final KNN

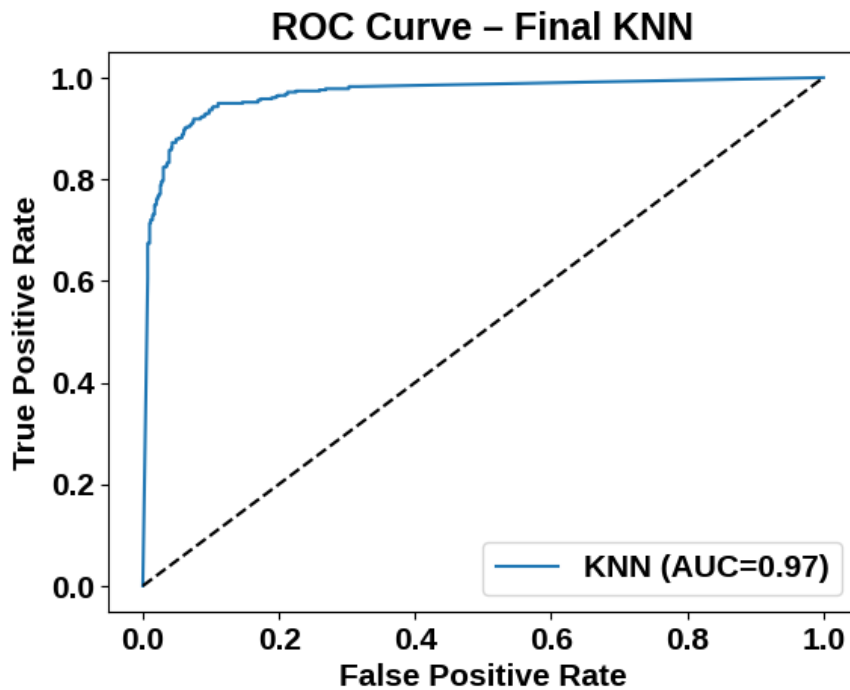


Figure 10: ROC for Final KNN

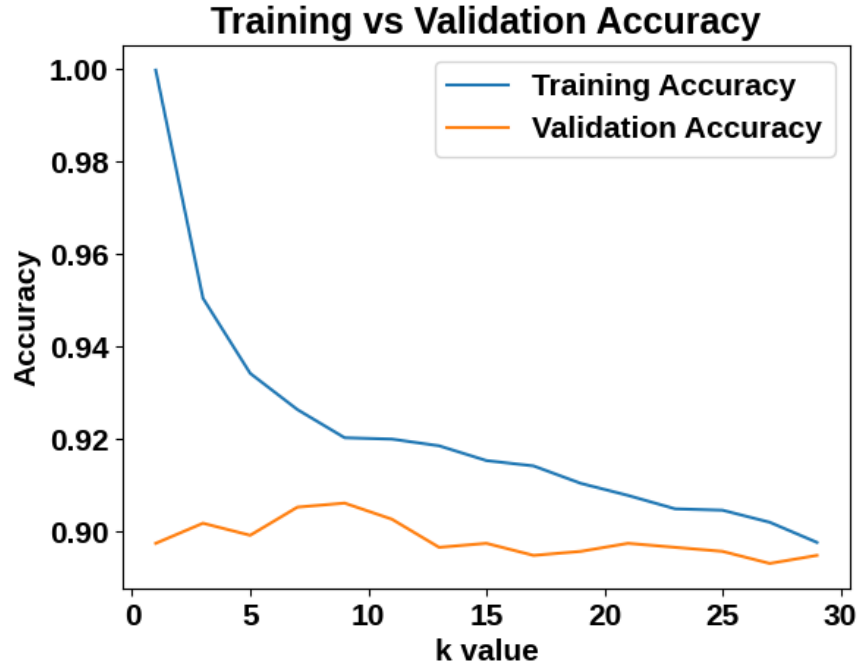


Figure 11: Training vs Validation Accuracy for KNN

Performance Tables

Table 1: Naïve Bayes Performance Metrics

Metric	Gaussian NB	Multinomial NB	Bernoulli NB
Accuracy	0.8253	0.7767	0.8818
Precision	0.7077	0.7244	0.8767
Recall	0.9493	0.7004	0.8149
F1 Score	0.8109	0.7122	0.8447
Specificity	0.7446	0.8263	0.9253
Training Time (s)	0.0085	0.01231	0.0201

Table 2: KNN Hyperparameter Tuning

Search Method	Best Parameters	Best CV Accuracy
Grid Search	k=13, distance, KDTree	0.92405
Randomized Search	k=6, distance, BallTree	0.9249

Table 3: KNN Performance using KDTree

Metric	Value
Optimal k	6
Accuracy	0.9209
Precision	0.9172
Recall	0.8788
F1 Score	0.8976
Training Time (s)	0.0073
Prediction Time (s)	0.6574

Table 4: KNN Performance using BallTree

Metric	Value
Optimal k	6
Accuracy	0.9209
Precision	0.9172
Recall	0.8788
F1 Score	0.8976
Training Time (s)	0.0073
Prediction Time (s)	0.6574

Table 5: Comparison of Neighbor Search Algorithms

Criterion	KDTree	BallTree
Accuracy	0.9209	0.9209
Training Time (s)	0.0073	0.0073
Prediction Time (s)	0.6574	0.6574
Memory Usage	Low / Medium	Medium / High

Overfitting and Underfitting Analysis

- Very small values of the parameter k caused the model to memorize training samples, resulting in high training accuracy but poor performance on unseen data.
- Very large values of k led to excessive smoothing, preventing the model from learning important patterns and increasing prediction errors.
- An intermediate range of k values provided a better trade-off between bias and variance.
- Hyperparameter optimization techniques improved the model's ability to generalize to new data.
- The analysis highlights the importance of selecting appropriate model complexity to avoid both overfitting and underfitting.
- Naive Bayes tends to introduce greater bias as it assumes conditional independence among features, which can limit model flexibility.
- The K-Nearest Neighbors (KNN) algorithm demonstrates increased variance when smaller values of k are used, making it sensitive to noise in the training data.

- Careful selection and tuning of hyperparameters helps achieve a more balanced trade-off between bias and variance, leading to improved generalization performance.

Observations and Learning Outcomes

The Naive Bayes models were efficient in terms of computation and training time, whereas the optimized KNN classifiers delivered superior predictive accuracy. Among the neighbor search strategies, BallTree showed improved computational performance compared to KDTree. Overall, effective data preprocessing, exploratory visualization, and systematic hyperparameter optimization played a crucial role in enhancing model accuracy and generalization capability.